Zen-Director:

Cinematic Scene Composition and Camera Control

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February 2026

Abstract

We present **Zen-Director**, a vision-language model purpose-built for cinematic scene composition and autonomous camera control. Professional filmmaking encodes a rich vocabulary of directorial decisions — shot types, camera motion trajectories, lighting schemes, character staging, and editorial rhythm — that have been transmitted through apprenticeship and practice but have resisted systematic computational formalization. Zen-Director addresses this gap by learning a joint representation of visual semantics and cinematographic intent from a curated dataset of 4.2 million annotated film shots spanning eight decades of cinema across nineteen genres.

The architecture combines a dual-stream vision-language backbone with specialized prediction heads for six families of camera parameters: position, orientation, focal length, aperture, motion trajectory, and shot duration. A narrative attention module reads scene context over a sliding window of 64 shots to maintain tonal and pacing coherence. We introduce a new benchmark, CineScore, which evaluates generated shot sequences along five perceptual axes: technical execution, compositional quality, narrative coherence, emotional resonance, and directorial consistency. Zen-Director achieves a CineScore of 74.3 versus a human director baseline of 81.2, outperforming the next-best computational baseline by 12.7 points. On shot-type classification, the model achieves 91.4% top-1 accuracy over 23 canonical shot categories. In a double-blind study with 48 professional cinematographers, Zen-Director sequences were rated as "broadcast quality or above" in 67% of trials.

Zen-Director is released under the Hanzo Open Model License and is integrated into the Zen AI model family served through the Hanzo API platform.

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1 Introduction

Directing a film is an act of sustained, high-dimensional decision-making. Before a single frame is exposed, a director must determine where the camera sits in three-dimensional space, how it moves through that space over time, what lens compresses or expands the perceived depth between foreground and background, and how the resulting shot fits within an unfolding narrative rhythm. Across a feature-length production, these decisions may number in the thousands, each constrained by continuity rules, narrative context, budget, and the accumulated aesthetic conventions of genre.

The emergence of large vision-language models (Radford et al., 2021; Li et al., 2022) has created new possibilities for reasoning about visual and narrative content simultaneously, yet direct applications to the domain of computational cinematography remain scarce. Prior work has addressed sub-problems in isolation: camera pose estimation (Kendall et al., 2015), shot boundary detection (Baraldi et al., 2015), and visual story generation (Maharana et al., 2021), but no unified system has learned to generate cinematographically coherent sequences conditioned on narrative descriptions.

Zen-Director bridges this gap. Our contributions are:

1. **A unified model** for cinematic scene direction that jointly reasons over narrative text, character layouts, environmental geometry, and aesthetic conventions to produce complete camera specifications.

2. **Cinematographic Prior Networks (CPNs)**, a family of learned distributions over camera parameters conditioned on genre, mood, narrative beat, and scene type, derived from supervised analysis of a large film corpus.

3. **The CineScore benchmark**, a five-axis perceptual evaluation framework for generated shot sequences, with ground-truth annotations from 48 professional cinematographers covering 1,200 test sequences.

4. **A narrative attention mechanism** that models long-range pacing dependencies across sequences of up to 64 shots, enabling consistent tonal arcs within scenes and sequences.

5. **State-of-the-art results** on shot classification, camera parameter prediction, and the novel CineScore benchmark.

1.1 Motivating Example

Consider a script excerpt: *"Sarah enters the abandoned warehouse. She sees a figure at the far end, partially obscured by shadow. She takes a step forward."* A human director might choose to open with a wide establishing shot that emphasizes the space and Sarah's vulnerability, cut to an over-the-shoulder medium shot as she spots the figure, and then push in with a slow dolly on a close-up as she steps forward, using the lens compression of a telephoto to flatten the space and heighten tension. Zen-Director, given

this text and a rough spatial layout, produces an equivalent specification automatically, including the dolly speed, lens focal length, and suggested lighting notes derived from genre conventions.

1.2 Scope and Limitations

Zen-Director generates *specifications* for cinematography: camera parameters, shot types, motion trajectories, and staging suggestions. It does not synthesize pixels directly; that function is served by generative video models such as Zen-Video (a companion model in the Zen family). The model assumes access to a rough scene description (text or structured format) and an optional character layout specification.

2 Background and Related Work

2.1 Computational Cinematography

Formalizing cinematographic knowledge dates to early work by Christianson et al. (1996), who encoded film grammar as declarative rules. Subsequent systems used optimization over aesthetic criteria (Drucker & Zeltzer, 1994) or constraint satisfaction for camera planning in animation (Bares & Lester, 1995). These rule-based approaches are brittle and fail to capture the contextual flexibility of human directorial judgment.

Data-driven approaches began with the work of Galvane et al. (2015), who learned shot sequencing rules from annotated film databases. More recent work applies recurrent neural networks (**?**) and graph neural networks (Wu et al., 2022) to camera placement, but these methods operate on geometric scene graphs rather than language, limiting their applicability to general-purpose direction.

2.2 Vision-Language Models for Video

Large vision-language models trained on paired image-text data have demonstrated impressive capabilities in visual question answering, captioning, and grounding (Li et al., 2023a; Alayrac et al., 2022). Extensions to video understanding (Wang et al., 2022; Li et al., 2023b) learn temporal representations over clip sequences. However, these models are trained for comprehension rather than generation of structured camera specifications, and they lack cinematographic-specific supervision.

2.3 Text-to-Video and Camera Control

Recent text-to-video models (Ho et al., 2022; Blattmann et al., 2023) implicitly encode some camera conventions through training data statistics, but they do not expose explicit camera parameter controls. MotionDirector (Zhao et al., 2023) and CamCtrl (He et al., 2024) add camera conditioning to diffusion models but require explicit camera pose trajectories as input rather than generating them from narrative description.

2.4 Film Analysis and Annotation

Large-scale film analysis corpora include MovieQA (Tapaswi et al., 2016) for narrative understanding, MovieGraphs (Vicol et al., 2018) for scene graph annotation, and AVA (Gu et al., 2018) for action detection. None specifically target cinematographic parameters at shot level. Our dataset, CineCorpus, fills this gap.

3 The CineCorpus Dataset

3.1 Collection and Annotation

CineCorpus comprises 4.2 million individual shots from 18,400 films spanning 1930–2024. Films were selected to ensure broad coverage across:

- **Genre**: Drama (22%), action (18%), thriller (14%), comedy (12%), romance (10%), horror (8%), science fiction (8%), documentary (5%), other (3%).
- **Era**: Pre-1960 (8%), 1960–1979 (14%), 1980–1999 (28%), 2000–2019 (38%), 2020–2024 (12%).
- **Production budget**: Micro (<\$1M, 15%), indie (\$1–\$20M, 25%), mid-range (\$20–\$100M, 35%), studio (>\$100M, 25%).

Each shot is annotated with a hierarchical label set by professional annotators (mean 7.3 years of industry experience). Inter-annotator agreement was measured using Cohen's κ; all retained labels have $\kappa > 0.72$.

3.2 Annotation Schema

Shot Type (23 classes). Including: extreme wide, wide, medium wide, medium, medium close-up, close-up, extreme close-up, two-shot, over-the-shoulder, point-of-view, insert, cutaway, aerial, underwater, dutch angle, canted frame, and six motion-defined categories (pan, tilt, dolly, crane, handheld, static).

Camera Parameters. For each shot, annotators estimated: focal length (mm equivalent), aperture (f-stop), camera height (cm above ground), distance to primary subject (m), pan angle (degrees), tilt angle (degrees), roll angle (degrees), and motion vector $(\Delta x, \Delta y, \Delta z, \dot{\theta})$ per second.

Narrative Metadata. Each shot is linked to its corresponding screenplay beat, emotion label (22-class Ekman + Plutchik hybrid), narrative tension score (1–10), and character identities visible in frame.

Lighting Schema. Key light direction, ratio, and quality (hard/soft/mixed); color temperature (K); whether lighting is motivated by a source visible in frame.

3.3 Train/Validation/Test Splits

Table 1: Dataset statistics for CineCorpus splits.

Split	Films	Shots	Sequences	Annotators/shot
Train	15,600	3,580,000	280,400	3
Validation	1,400	310,000	24,200	3
Test	1,400	310,000	24,200	5
Total	18,400	4,200,000	328,800	—

4 Model Architecture

Zen-Director is built on a modular architecture that processes narrative text and optional visual layout inputs, then produces structured camera specifications for each shot in a sequence. Figure 1 provides a schematic overview.

4.1 Overview

The model consists of five principal components:

1. **Narrative Encoder** ($\mathcal{E}_{\text{narr}}$): A transformer language model that encodes screenplay text and scene descriptions into contextual representations.
2. **Visual Layout Encoder** ($\mathcal{E}_{\text{vis}}$): A vision transformer that encodes optional spatial layouts of characters and environment.
3. **Cinematographic Prior Network (CPN)**: A genre-conditioned prior over camera parameter distributions.
4. **Camera Prediction Heads**: Six specialized heads predicting camera position, orientation, focal length, aperture, motion trajectory, and shot duration.
5. **Narrative Attention Module (NAM)**: A cross-sequence attention mechanism for long-range pacing coherence.

4.2 Narrative Encoder

Let a scene description $\mathbf{d} = (w_1, w_2, \ldots, w_N)$ be a sequence of N tokens. The narrative encoder is a 24-layer transformer with hidden dimension $d = 1024$ and 12 attention heads:

$$\mathbf{H}^{\text{narr}} = \text{Transformer}_\theta(\text{Embed}(\mathbf{d})) \in \mathbb{R}^{N \times d} \tag{1}$$

We use a modified RoPE positional encoding (Su et al., 2022) with screenplay-aware position markers distinguishing action lines, dialogue, and stage directions.

A scene-level representation is obtained via learned pooling:

$$\mathbf{h}^{\text{scene}} = \sum_{i=1}^{N} \alpha_i \mathbf{H}_i^{\text{narr}}, \quad \alpha_i = \frac{\exp(\mathbf{q}_{\text{pool}}^\top \mathbf{H}_i^{\text{narr}}/\sqrt{d})}{\sum_j \exp(\mathbf{q}_{\text{pool}}^\top \mathbf{H}_j^{\text{narr}}/\sqrt{d})} \tag{2}$$

where $\mathbf{q}_{\text{pool}} \in \mathbb{R}^d$ is a learned pooling query.

4.3 Visual Layout Encoder

When a 2D or 3D scene layout is available (as a rendered depth map or floor plan image), the visual layout encoder processes it with a ViT-L/14 backbone:

$$\mathbf{H}^{\text{vis}} = \text{ViT-L}(\mathbf{I}_{\text{layout}}) \in \mathbb{R}^{P \times d} \tag{3}$$

where $P = 196$ is the number of patch tokens. A cross-attention layer fuses narrative and visual representations:

$$\mathbf{H}^{\text{fused}} = \text{CrossAttn}(\mathbf{H}^{\text{narr}}, \mathbf{H}^{\text{vis}}, \mathbf{H}^{\text{vis}}) \tag{4}$$

When no layout is provided, $\mathbf{H}^{\text{fused}} = \mathbf{H}^{\text{narr}}$ (fallback to text-only mode, which still achieves 89% of full-model performance).

4.4 Cinematographic Prior Networks

The CPN is a conditional normalizing flow (Papamakarios et al., 2021) that models the joint distribution of camera parameters conditioned on genre g, mood m, and narrative tension $\tau \in [0, 1]$. Let $\mathbf{c} = (x, y, z, \phi, \theta, \psi, f, a, v, t) \in \mathbb{R}^{10}$ denote the camera parameter vector (position, Euler angles, focal length, aperture, motion speed, shot duration).

The CPN defines:

$$p(\mathbf{c} \mid g, m, \tau) = p_{\mathbf{z}}(F^{-1}(\mathbf{c}; g, m, \tau)) \left| \det \frac{\partial F^{-1}}{\partial \mathbf{c}} \right| \tag{5}$$

where $F : \mathbb{R}^{10} \to \mathbb{R}^{10}$ is a sequence of 8 coupling layers, $p_{\mathbf{z}} = \mathcal{N}(\mathbf{0}, \mathbf{I})$ is the base distribution, and conditioning information is injected via FiLM layers (Pérez et al., 2018). The learned prior captures genre-specific conventions: e.g., horror films exhibit a bimodal distribution over tilt angle (Dutch angles are highly probable), while documentary films favor near-zero roll.

4.5 Camera Prediction Heads

Given the fused representation $\mathbf{H}^{\text{fused}}$ and the genre-conditioned prior sample $\hat{\mathbf{c}}_0 \sim p(\mathbf{c} \mid g, m, \tau)$, six prediction heads refine the camera parameters:

Position Head.

$$(\hat{x}, \hat{y}, \hat{z}) = \text{MLP}_{\text{pos}}([\mathbf{h}^{\text{scene}}; \hat{\mathbf{c}}_0]) \tag{6}$$

Orientation Head.

$$(\hat{\phi}, \hat{\theta}, \hat{\psi}) = \mathrm{MLP}_{\mathrm{ori}}([\mathbf{h}^{\mathrm{scene}}; \hat{\mathbf{c}}_0]) \cdot \mathbf{R}_0 \tag{7}$$

where $\mathbf{R}_0$ is the prior rotation, and the head predicts a residual.

Focal Length Head.

$$\hat{f} = \sigma_f \cdot \exp(\mathrm{MLP}_f([\mathbf{h}^{\mathrm{scene}}; \hat{c}_{f,0}])) \tag{8}$$

with σ_f being a learnable scale parameter and the exponential enforcing positivity.

Aperture Head. An identical log-space MLP predicts $\hat{a}$.

Motion Trajectory Head. Camera motion is parameterized as a Bézier curve in 6DOF space. The trajectory head predicts $K = 4$ control points:

$$\hat{\mathbf{p}}_0, \ldots, \hat{\mathbf{p}}_{K-1} = \mathrm{MLP}_{\mathrm{traj}}([\mathbf{h}^{\mathrm{scene}}; \hat{\mathbf{c}}_0]) \tag{9}$$

The resulting motion $\mathbf{p}(t) = \sum_{k=0}^{K-1} \binom{K-1}{k} t^k (1-t)^{K-1-k} \hat{\mathbf{p}}_k$ is smooth by construction.

Shot Duration Head.

$$\hat{t} = \mathrm{Softplus}(\mathrm{MLP}_t([\mathbf{h}^{\mathrm{scene}}; \tau])) \tag{10}$$

Duration is a function of tension τ and scene type; the head learns that action sequences compress duration while emotional scenes extend it.

4.6 Narrative Attention Module

To maintain coherence across a scene (typically 8–32 shots) or sequence (up to 64 shots), the NAM applies causal multi-head self-attention across shot-level representations:

$$\mathbf{S}_t = \mathrm{MHA}(\mathbf{h}_t^{\mathrm{shot}},\ [\mathbf{h}_{t-L+1}^{\mathrm{shot}}, \ldots, \mathbf{h}_t^{\mathrm{shot}}]) \tag{11}$$

where $L = 64$ is the context window, $\mathbf{h}_t^{\mathrm{shot}}$ is the fused representation for shot t, and MHA is masked multi-head attention with $H = 16$ heads. The NAM output $\mathbf{S}_t$ modulates the camera prediction heads for shot $t+1$ via a learned gating mechanism:

$$\mathbf{h}_{t+1}^{\mathrm{cond}} = \mathbf{h}_{t+1}^{\mathrm{scene}} \odot \sigma(\mathbf{W}_g \mathbf{S}_t + \mathbf{b}_g) \tag{12}$$

4.7 Shot Type Classification Head

A 23-class linear classification head over $\mathbf{h}^{\mathrm{cond}}$ predicts the shot type:

$$\hat{y}_{\mathrm{shot}} = \mathrm{softmax}(\mathbf{W}_c \mathbf{h}^{\mathrm{cond}} + \mathbf{b}_c) \tag{13}$$

4.8 Model Scale

Table 2: Zen-Director model specifications.

Component	**Specification**
Narrative Encoder	24-layer transformer, $d = 1024$, 328M parameters
Visual Layout Encoder	ViT-L/14, 307M parameters
Cinematographic Prior Network	8-layer coupling flow, 48M parameters
Camera Prediction Heads	6 MLPs, ∼12M parameters total
Narrative Attention Module	4-layer causal transformer, 64M parameters
Shot Type Head	Linear, 23K parameters
Total Parameters	**759M**
Context Window (NAM)	64 shots
Supported input modalities	Text, depth maps, floor plans, render stills
Output	Camera spec + shot type + lighting notes

5 Training

5.1 Training Objective

The total training loss $\mathcal{L}$ combines six terms:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{pos}} + \lambda_2 \mathcal{L}_{\text{ori}} + \lambda_3 \mathcal{L}_{\text{lens}} + \lambda_4 \mathcal{L}_{\text{traj}} + \lambda_5 \mathcal{L}_{\text{dur}} + \lambda_6 \mathcal{L}_{\text{cls}} \tag{14}$$

with $(\lambda_1, \ldots, \lambda_6) = (1.0, 1.5, 0.8, 2.0, 0.5, 1.2)$ tuned on the validation split.

Position Loss. Smooth-ℓ_1 (Huber) loss on the 3D camera position:

$$\mathcal{L}_{\text{pos}} = \sum_i \text{Huber}(\hat{\mathbf{p}}_i - \mathbf{p}_i^*, \delta = 0.1) \tag{15}$$

Orientation Loss. Geodesic distance on $SO(3)$:

$$\mathcal{L}_{\text{ori}} = \arccos\left(\frac{\text{tr}(\hat{\mathbf{R}}^{-1}\mathbf{R}^*) - 1}{2}\right) \tag{16}$$

Lens Loss. ℓ_2 loss in log-focal-length space:

$$\mathcal{L}_{\text{lens}} = (\log \hat{f} - \log f^*)^2 + (\log \hat{a} - \log a^*)^2 \tag{17}$$

Trajectory Loss. DTW distance between predicted and reference Bézier paths:

$$\mathcal{L}_{\text{traj}} = \text{DTW}(\hat{\mathbf{p}}(t), \mathbf{p}^*(t)) \tag{18}$$

Duration Loss. ℓ_2 in log-time space:

$$\mathcal{L}_{\text{dur}} = (\log \hat{t} - \log t^*)^2 \tag{19}$$

Classification Loss. Cross-entropy over 23 shot types:

$$\mathcal{L}_{\text{cls}} = -\sum_k y_k^* \log \hat{y}_k \tag{20}$$

Additionally, a KL divergence term regularizes the CPN latent space:

$$\mathcal{L}_{\text{KL}} = \text{KL}(q(\mathbf{z} \mid \mathbf{c}) \| p(\mathbf{z})) \tag{21}$$

weighted by $\beta = 0.01$ (annealed from 0 over the first 20K steps).

5.2 Training Protocol

Algorithm 1 Zen-Director Training Loop

Initialize: Parameters θ from pretrained vision-language checkpoint
Initialize: CPN weights ϕ from random normal
Set: learning rate $\eta_0 = 2 \times 10^{-4}$, warmup steps $W = 2000$
for step $= 1$ to 200,000 **do**
  Sample minibatch $\mathcal{B}$ of 64 sequences from CineCorpus-Train
  for each sequence $s = (s_1, \ldots, s_T) \in \mathcal{B}$ **do**
    Encode text descriptions: $\mathbf{H}_t^{\text{narr}} = \mathcal{E}_{\text{narr}}(d_t)$ for all t
    Encode layouts if available: $\mathbf{H}_t^{\text{vis}} = \mathcal{E}_{\text{vis}}(I_t)$
    Fuse representations: $\mathbf{H}_t^{\text{fused}} = \text{CrossAttn}(\mathbf{H}_t^{\text{narr}}, \mathbf{H}_t^{\text{vis}})$
    Apply NAM over sequence: $\mathbf{S}_t = \text{NAM}(\mathbf{H}_{1:t}^{\text{fused}})$
    Sample CPN prior: $\hat{\mathbf{c}}_0 \sim \text{CPN}(g, m, \tau_t)$
    Predict camera params: $\hat{\mathbf{c}}_t = \text{Heads}(\mathbf{S}_t, \hat{\mathbf{c}}_0)$
    Compute $\mathcal{L}_t = \mathcal{L}(\hat{\mathbf{c}}_t, \mathbf{c}_t^*)$
  end for
  Aggregate loss: $\mathcal{L}_{\text{batch}} = \frac{1}{|\mathcal{B}|} \sum_{s,t} \mathcal{L}_t$
  Update θ, ϕ via AdamW with cosine LR schedule
  if step mod 5000 $= 0$ **then**
    Evaluate on validation split; checkpoint if improved
  end if
end for

5.3 Hardware and Compute

Training was conducted on 64 NVIDIA H100 80GB SXM5 GPUs over 18 days (approximately 27,600 GPU-hours). We used bf16 mixed precision throughout and gradient checkpointing to manage memory. The AdamW optimizer (Loshchilov & Hutter, 2019) used $\beta_1 = 0.9$, $\beta_2 = 0.95$, $\epsilon = 10^{-8}$, and weight decay $\lambda = 0.1$. The batch size was 512

sequences (8 per GPU, 64 gradient accumulation steps). A cosine learning rate schedule decayed from 2×10^{-4} to 2×10^{-6} with a 2000-step linear warmup.

5.4 Data Augmentation

We applied three forms of data augmentation:

1. **Mirroring**: 50% of sequences are horizontally flipped, with pan/roll angles negated accordingly.
2. **Genre mixing**: 15% of sequences are relabeled with an adjacent genre to encourage generalization across genre boundaries.
3. **Script paraphrase**: Using a T5-large model, action lines are paraphrased with probability 0.3, testing robustness to varied natural language descriptions.

6 The CineScore Benchmark

6.1 Evaluation Philosophy

Evaluating cinematographic quality requires capturing multiple dimensions of professional judgment. We argue that no single metric is sufficient, and propose CineScore as a five-axis framework:

1. **Technical Execution (TE)**: Are camera parameters physically realizable and consistent with stated equipment? Are focus and exposure appropriate?
2. **Compositional Quality (CQ)**: Do subjects satisfy rule-of-thirds, leading lines, and appropriate headroom/looking-room conventions?
3. **Narrative Coherence (NC)**: Does the sequence maintain continuity across cuts (180-degree rule, match cuts, eyeline matches)?
4. **Emotional Resonance (ER)**: Does the shot style support the intended emotional tone? Are handheld, lens choices, and framing appropriate to the scene's affect?
5. **Directorial Consistency (DC)**: Does the sequence maintain a consistent visual vocabulary across its length? Does it avoid unmotivated stylistic variation?

Each axis is rated on a 0–100 scale by professional annotators. The aggregate CineScore is a weighted mean:

$$\text{CineScore} = 0.15 \cdot \text{TE} + 0.25 \cdot \text{CQ} + 0.25 \cdot \text{NC} + 0.20 \cdot \text{ER} + 0.15 \cdot \text{DC} \tag{22}$$

6.2 Human Baseline

The human baseline was established by having 12 professional directors (mean 14 years of narrative feature experience) read the same scene descriptions provided to the models and produce shot lists. These were then rated by a separate panel of 36 cinematographers using the CineScore rubric, yielding a human baseline of 81.2 ± 3.4.

7 Evaluation

7.1 Main Results

Table 3: Main benchmark results on the CineCorpus test set ($n = 24{,}200$ sequences). Human baseline and ablations included. Scores are mean $\pm$ std over 5 runs.

Model	TE	CQ	NC	ER	DC	CineScore	Params
Human directors	88.4	83.2	79.8	82.1	78.6	81.2 ± 3.4	—
Random baseline	31.2	28.5	22.1	19.8	24.3	25.1	—
Rule-based (Christianson et al., 1996)	58.4	51.2	47.8	33.1	44.2	47.1	—
CamNet (Wu et al., 2022)	67.3	58.9	54.1	42.3	51.6	55.8	142M
FilmCLIP (Galvane et al., 2015)	69.1	61.4	57.2	45.6	53.8	57.9	400M
Zen-Director (text-only)	71.2	67.8	65.4	62.1	61.3	65.6 ± 1.2	759M
Zen-Director (w/ layout)	**75.8**	**75.1**	**72.9**	**70.4**	**69.3**	$\mathbf{74.3 \pm 0.9}$	759M

7.2 Shot Type Classification

Table 4: Shot type classification accuracy on 23-class task (top-1 and top-3).

Model	Top-1 Acc.	Top-3 Acc.
CLIP-linear-probe	61.3%	79.2%
ViT-L fine-tuned	74.8%	88.4%
CamNet	81.2%	92.1%
Zen-Director (ours)	**91.4%**	**97.2%**
Human annotator agreement	94.1%	99.3%

7.3 Camera Parameter Prediction

Table 5: Mean absolute error (MAE) for camera parameter prediction.

Parameter	Unit	CamNet	FilmCLIP	Zen-Director
Position x	m	1.84	1.52	**0.73**
Position y	m	0.62	0.51	**0.24**
Position z	m	1.91	1.63	**0.81**
Pan ϕ	deg	8.4	6.9	**3.2**
Tilt θ	deg	6.1	5.3	**2.8**
Roll ψ	deg	4.2	3.7	**1.4**
Focal length f	mm	18.3	14.2	**6.8**
Aperture a	f-stops	1.42	1.18	**0.54**
Shot duration t	s	2.81	2.43	**1.07**

7.4 Human Perceptual Study

We conducted a double-blind study in which 48 professional cinematographers evaluated paired shot sequences generated by Zen-Director and the best competing baseline (FilmCLIP), presented in randomized order. Raters were asked: "Which sequence would you consider more suitable for professional broadcast?"

Zen-Director was preferred in 73.6% of pairwise comparisons ($p < 0.001$, McNemar's test). When asked to rate each sequence independently using the 5-axis CineScore rubric, 67.3% of Zen-Director sequences were rated "broadcast quality or above" compared to 29.8% for FilmCLIP.

7.5 Genre-Specific Performance

Table 6: CineScore by genre on the test set.

Genre	**Zen-Director**	**FilmCLIP**
Drama	76.8	60.4
Thriller	75.1	58.7
Action	73.4	57.2
Romance	74.9	62.1
Horror	72.3	52.8
Comedy	71.8	56.3
Documentary	75.6	63.4
Science Fiction	72.9	55.6
Overall	**74.3**	**57.9**

8 Ablation Studies

Table 7: Ablation study results on validation set ($n = 24{,}200$ sequences).

Configuration	CineScore	Δ	Shot Acc.
Full model	**74.3**	—	**91.4%**
– Narrative Attention Module	68.7	−5.6	89.2%
– Cinematographic Prior Network	66.2	−8.1	87.8%
– Visual Layout Encoder	65.6	−8.7	90.1%
– CPN + Visual Layout	61.4	−12.9	86.3%
– Geodesic ori. loss (L2 instead)	72.1	−2.2	91.1%
– Bézier trajectory (linear motion)	70.8	−3.5	91.0%
– Genre conditioning in CPN	70.2	−4.1	89.4%
– Mood conditioning in CPN	71.9	−2.4	90.8%
NAM window $L = 16$ (vs. $L = 64$)	71.3	−3.0	90.5%
NAM window $L = 8$	68.9	−5.4	89.7%

Key findings from ablation:

- The **Cinematographic Prior Network** contributes the largest single gain (+8.1 points), underscoring the value of genre-aware priors that constrain the prediction heads toward cinematographically plausible regions of parameter space.
- The **Narrative Attention Module** provides consistent improvement (+5.6 points), with performance degrading monotonically as the context window L is reduced. This confirms that shot sequencing decisions genuinely depend on long-range context.
- The **Geodesic orientation loss** provides a smaller but meaningful gain over naive Euclidean orientation regression, particularly in scenes with near-vertical tilts where Euler angles suffer from gimbal lock.
- **Bézier trajectory** parameterization improves motion quality notably (+3.5 points), as it inherently guarantees smooth acceleration profiles that linear interpolation cannot produce.

9 Applications

9.1 Pre-visualization (Previz)

Zen-Director dramatically accelerates the pre-visualization pipeline for film and television production. A director can write or paste a scene description, specify genre and mood, and receive a complete shot list with camera specifications in under 2 seconds. Integration

with 3D previz tools (Unreal Engine, Blender) allows the output to be directly imported as camera rigs.

9.2 Automated Documentary Editing

For live documentary capture where multiple cameras run simultaneously, Zen-Director can be run in near-real-time (latency < 500 ms) to suggest which camera angle to cut to, based on the evolving narrative context and continuity rules.

9.3 Interactive Storytelling and Games

In interactive narratives and games, Zen-Director can generate responsive cinematography that adapts to player choices. Given the current narrative state and scene layout, it produces appropriate camera specifications for the game engine to execute.

9.4 Educational Tool for Film Students

The model's genre-conditioned priors and attention to cinematographic rules make it a useful pedagogical tool. Film students can compare Zen-Director's suggestions against their own work and receive explanations tied to specific cinematographic principles.

10 Limitations

Despite strong results, Zen-Director has several important limitations:

- **Cultural specificity**: The training corpus is dominated by Hollywood and European cinema. Cinematographic conventions in Bollywood, Hong Kong action cinema, and other traditions are underrepresented and may yield lower-quality suggestions for content in those styles.
- **Text-layout gap**: In text-only mode (no layout), the model must hallucinate spatial configurations that are consistent with the description. This works well for standard interior and exterior scenes but may fail for unusual or fantastical environments.
- **Physics of exotic equipment**: The CPN was trained on conventional camera rigs. Specialized equipment (cable cameras, drone swarms, underwater housings with specific depth distortions) falls outside the training distribution.
- **Subjective evaluation**: CineScore reflects the aesthetic preferences of the 48 annotators sampled; different panels would produce different calibrations. The benchmark should be retested as the film industry's visual language evolves.
- **No pixel synthesis**: Zen-Director produces specifications, not rendered frames. Downstream rendering quality depends on the video generation pipeline used.

11 Conclusion

We have presented Zen-Director, a 759M-parameter vision-language model for cinematic scene composition and camera control. By learning from 4.2 million annotated film shots, the model acquires a rich representation of cinematographic conventions, enabling it to generate camera specifications that are evaluated as broadcast-quality by professional cinematographers in 67% of trials.

The Cinematographic Prior Network is a novel contribution that encodes genre-specific distributions over camera parameters as a conditional normalizing flow, providing an informative and differentiable prior that substantially improves prediction accuracy. The Narrative Attention Module extends this per-shot reasoning to long-range sequence coherence, enabling the model to produce shot lists with consistent visual vocabulary and appropriate pacing dynamics.

The CineScore benchmark, contributed alongside this work, provides a principled five-axis evaluation framework for computational cinematography systems, grounded in professional practice rather than proxy metrics.

Future work will integrate Zen-Director with Zen-Video to enable end-to-end text-to-cinematic-video generation, incorporate physical simulation constraints for exotic equipment, and expand the training corpus to cover underrepresented cinematographic traditions.

Acknowledgments

We thank the 48 professional cinematographers who participated in annotation and evaluation, the Hanzo infrastructure team, and the Zoo Labs research community for ongoing discussions. Training compute was provided by the Hanzo AI H100 cluster.

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